Assignment 2: supplementary note

Multiscale Modelling and Computation (APL410/AML7200)

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1 Lattices and potentials

Q1, Q2, Q4. Triangular lattice with the Lennard–Jones pair potential $\phi_{\mathrm{LJ}}(r) = 4(r^{-12} - r^{-6})$, truncated with the cutoff correction to $\phi(r)$. Each atom has neighbour shells at

$$a \text{ (6 atoms)}, \quad \sqrt{3}a \text{ (6)}, \quad 2a \text{ (6)}, \quad \sqrt{7}a \text{ (12)}, \ \ldots$$

With $a = 1.116153$ and $r_{\mathrm{cut}} = 2.5$, the first three shells interact. This a minimizes the energy of the perfect lattice for the cutoff-corrected potential. It is slightly smaller than the minimum of the pair potential: the nearest-neighbour bonds are slightly compressed ($\phi'(a) = -0.42$), and the more distant pairs, which are stretched, balance them.

Q3. The edge dislocation is easiest to see in a square lattice, where it appears as an extra half-column of atoms. Lennard–Jones alone cannot hold a square lattice. A central (pair) force acts along the line joining two atoms. Shear a square cell by a small angle γ: the nearest-neighbour bonds rotate but change length only at order γ^2, so they give no shear stiffness. The resistance to shear comes from the diagonal pairs at $\sqrt{2}a$, which are stretched or compressed at order γ. For Lennard–Jones, $\phi''_{\mathrm{LJ}}(r) < 0$ beyond $r \approx 1.24$, so the diagonal pairs *soften* the lattice in shear, and a square lattice collapses into a triangular arrangement. A Gaussian well centred at the diagonal distance gives the diagonal pairs a positive curvature:

$$\phi_{\mathrm{sq}}(r) = \phi_{\mathrm{LJ}}(r) - \epsilon_2 \exp\left[-\frac{(r - r_2)^2}{2s^2}\right], \qquad \epsilon_2 = 1,\ r_2 = 1.56,\ s = 0.1,$$

$$\phi'_{\mathrm{sq}}(r) = 24\left(r^{-7} - 2r^{-13}\right) + \epsilon_2 \frac{r - r_2}{s^2} \exp\left[-\frac{(r - r_2)^2}{2s^2}\right].$$

The well has curvature $\epsilon_2/s^2 = 100$ at r_2. The square-lattice spacing is $a = 1.103090$, and the well is centred at $r_2 = \sqrt{2}a$, so it stiffens the diagonal bonds without changing a. The square lattice then has energy -4.258 per atom, against -2.915 for the best triangular lattice. The cutoff correction is applied to ϕ_{sq} in the same way as to ϕ_{LJ}, and Algorithm 6.6 is unchanged because the potential is still a pair potential.

2 Formation energies

Definition. A defect is compared with a perfect crystal containing the same number of atoms. With $e_0 = E_N/N$ the energy per atom of the perfect crystal,

$$E_f^v = E_{N-1}^v - (N-1)\, e_0, \qquad E_f^i = E_{N+1}^i - (N+1)\, e_0.$$

For the vacancy, the removed atom is placed in the bulk of a perfect crystal elsewhere, where it has energy e_0. For the interstitial, the extra atom is taken from such a crystal.

Unrelaxed vacancy with a pair potential. The energy of one atom's bonds is $\sum_\beta \phi(r^{\alpha\beta}) = 2e_0$, because each bond is shared by two atoms. Removing the atom without relaxation deletes these bonds, so $E_{N-1}^v = E_N - 2e_0$ and

$$E_f^{v,\mathrm{unrelaxed}} = E_N - 2e_0 - (N-1)e_0 = -e_0,$$

the cohesive energy per atom. In real metals the vacancy formation energy is about a third of the cohesive energy (for Cu, about 1.3 against 3.5 eV). A pair potential cannot reproduce this, because

the strength of a bond does not depend on how many other bonds the atom has; many-body potentials (for example the embedded-atom method) are needed for that.

Relaxation. After the atom is removed, each neighbour carries an unbalanced force equal and opposite to the force the missing atom used to exert on it. A neighbour whose bond to the missing atom was compressed moves towards the vacant site; one whose bond was stretched moves away. The surrounding bonds are stiff, so the displacements u are small, and the energy gained is of order $\frac{1}{2}Fu$ per atom. The relaxation energy is therefore a small fraction of E_f^v.

Interstitial. An atom placed at the centre of a nearest-neighbour triangle is $a/\sqrt{3} \approx 0.64$ from three atoms, deep in the r^{-12} wall. The unrelaxed energy is very large, the forces are large, and relaxation pushes atoms apart over many spacings. Because the repulsive branch is very steep, compressing many bonds a little costs less than compressing a few bonds a lot, so the extra atom spreads its compression over as many bonds as it can. Even after relaxation many bonds remain compressed, so $E_f^i \gg E_f^v$. The same ordering holds in metals (in Cu, $E_f^i \approx 3$ eV, against about 1.3 eV for the vacancy).

Cell size. The displacements around a defect decay with distance. In a periodic cell the defect interacts with its periodic images through these displacement fields, so E_f depends on the cell size and converges as the cell grows. All calculations here are at fixed cell dimensions.

3 Visualizing displacements

- The displacement of atom α is $\mathbf{u}^\alpha = \mathbf{r}^\alpha_{\text{relaxed}} - \mathbf{r}^\alpha_{\text{initial}}$. With PBC, apply the nearest-image correction (Algorithm 6.8) to $\mathbf{u}^\alpha$, in case an atom has crossed the cell boundary.
- Draw an arrow from each initial position along $\mathbf{u}^\alpha$ (`quiver` in Matplotlib or MATLAB). Displacements around a vacancy are far smaller than a, so multiply every arrow by the same factor s (for example $s = 100$) and state s in the caption. Plot only a few neighbour shells around the defect.
- A second useful plot is the radial displacement $u_r = \mathbf{u}^\alpha \cdot (\mathbf{r}^\alpha - \mathbf{r}_d)/|\mathbf{r}^\alpha - \mathbf{r}_d|$ against the distance from the defect site $\mathbf{r}_d$. Its sign shows whether a shell moves towards ($u_r < 0$) or away from the defect.
- When atoms move by a sizeable fraction of a (interstitials, dislocations), arrows overlap. Plot the atoms instead, and join every pair closer than a bond cutoff with a line (first-neighbour bonds): $1.37a$ for the triangular lattice, $1.2a$ for the square lattice. In a perfect lattice this draws a regular triangular or square grid; a defect shows up as a break or distortion in the grid.

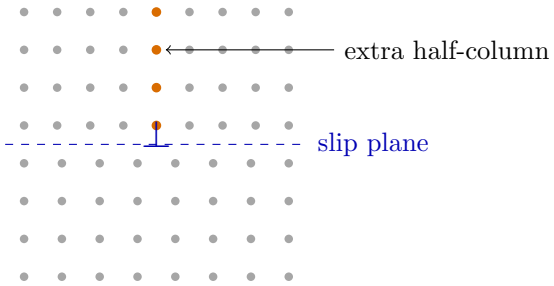
4 Coordination number

The coordination number n^α is the number of neighbours of atom α closer than r_c, with r_c between the first and second neighbour shells:

- triangular lattice: $r_c = 1.37a$ (shells at a and $1.73a$); a perfect-lattice atom has 6;
- square lattice: $r_c = 1.2a$ (shells at a and $1.41a$); a perfect-lattice atom has 4.

Near a defect, bonds are stretched or squeezed and some atoms have a different count. Atoms on a free surface also have fewer neighbours, so exclude a strip of about $2.5a$ next to each free surface, and the rows next to constrained groups, when looking for a defect inside a specimen. The count uses the same double loop over pairs as Algorithm 6.6: if $(r^{\alpha\beta})^2 < r_c^2$, add 1 to n^α and to n^β.

5 Edge dislocation



Geometry. The upper half of the crystal contains one more column of atoms than the lower half. This extra half-column ends at the **dislocation core**, just above the slip plane (symbol $\perp$). Far from the core the lattice is only bent elastically.

Burgers vector. Draw a closed circuit of lattice steps around the core (for example 6 steps right, 6 up, 6 left, 6 down). Repeat the same steps in a perfect lattice: the circuit fails to close, and the closure vector is the Burgers vector $\mathbf{b}$. Here $\mathbf{b} = (a, 0)$, which lies in the slip plane: that is what makes this an edge dislocation.

How the input was built (Volterra construction). A perfect block is cut along the half-plane $x = 0$, $y > 0$. The displacement field of an edge dislocation in an isotropic elastic medium (Poisson ratio ν),

$$u_x = \frac{b}{2\pi}\left[\theta + \frac{xy}{2(1-\nu)r^2}\right], \qquad u_y = -\frac{b}{2\pi}\left[\frac{1-2\nu}{4(1-\nu)}\ln\frac{r^2}{a^2} + \frac{x^2-y^2}{4(1-\nu)r^2}\right],$$

is applied to every atom, with the angle θ chosen to jump by 2π across the cut. The cut opens by b, and one extra column of atoms is placed in the gap. Elasticity is not valid within a few spacings of the core, so this configuration is only an initial guess and must be relaxed.

Glide. A shear stress τ on the slip plane exerts a force τb per unit length of dislocation line (Peach–Koehler force). The lattice resists the motion: the core must pass over an energy barrier to move by one spacing. In molecular statics there is no thermal activation, so the core moves only when τ is large enough to remove this barrier. That stress is the **Peierls stress**. Each time the core moves, the upper half of the crystal slips slightly over the lower half.

Free surfaces. A free surface carries no traction, so the stress field of the dislocation has to be cancelled there. The result is an attractive *image force* that pulls the dislocation towards the surface; it grows as the distance d to the surface decreases, roughly as $\mu b^2/[4\pi(1-\nu)d]$ (μ: shear modulus). When the dislocation reaches the surface it leaves the crystal. A surface step of height b remains, and the elastic energy of the dislocation is released.

6 Loading by prescribed displacement; reaction force

The lectures fix a constrained atom at a prescribed position, $\mathbf{r}^\alpha = \mathbf{R}^\alpha$ for $\alpha \in C_d$, and set its force to zero inside the minimizer. To load the specimen:

1. relax with the top and bottom groups fixed; call the relaxed top positions $\mathbf{X}^\alpha$;
2. for $u = \Delta u, 2\Delta u, \ldots$: set $\mathbf{R}^\alpha = \mathbf{X}^\alpha + u\,\mathbf{e}_x$ for the top atoms, start from the previous relaxed configuration, and minimize the mobile atoms.

Every recorded state is an energy minimum (**quasi-static** loading; no time, no inertia). The increment must be small. The minimizer finds the minimum nearest to its starting point, so a large increment can skip intermediate states, and it also needs more iterations.

Reaction. The loading device must hold the top group against the interatomic forces acting on it:

$$R_x = -\sum_{\alpha \in \text{top}} f_x^\alpha, \qquad f_x^\alpha = -\frac{\partial \mathcal{V}}{\partial x^\alpha}.$$

Use the forces *before* they are set to zero for constrained atoms. In the combined algorithm, that is after the pair loop and before the step $\mathbf{f}^\alpha := \mathbf{0}$, $\alpha \in C_d$.

Shear strain and stress. With H the distance between the top and bottom groups and W the length of the top group, the engineering shear strain is $\gamma = u/H$ and the average shear stress on the top boundary is $\tau = R_x/W$. In 2D the energy is per unit thickness, so τ is in units of ϵ/σ^2 (energy per unit area), not a 3D stress in Pa.

7 Implementation details

- **Nearest image and the neighbour list.** The nearest-image rule gives the correct distance only if the interaction range is at most half the cell length. With a Verlet list the range is r_{neigh}, so check $r_{\text{neigh}} \le L/2$. Here $r_{\text{neigh}} = 3.0$ and the smallest cell length is $L_y = 9.67$.
- **Rebuild test.** Compute each atom's displacement since the last build, using the nearest-image correction, and take the two largest magnitudes $\delta_{\max,1}, \delta_{\max,2}$. Store the positions at every build.
- **Counting work.** Count every pair distance you compute, both in the energy/force loop and while building lists. The N^2 search computes $N(N-1)/2$ distances per evaluation. The Verlet list computes the list length per evaluation, plus $N(N-1)/2$ per build.
- **Padding.** The list length grows with $(1+\epsilon_{\text{neigh}})^2$, while the number of builds falls as ϵ_{neigh} grows; the cheapest choice depends on how far the atoms move.
- **Timing.** Measure CPU time of the minimization only (for example `time.process_time()` in Python, `cputime` in MATLAB), not file input/output or plotting.
- **Stopping test and round-off.** The minimization stops when $\max_\alpha \|\mathbf{f}^\alpha\|$, the largest force magnitude over all mobile atoms, falls below the tolerance. Near convergence the energy decrease per step is roughly the square of the force divided by a stiffness, about 10^{-12} for forces of 10^{-5}. The total energy is of order 10^3 and double precision resolves it only to about 10^{-13}–10^{-12}. If the tolerance is too small, the sufficient-decrease test of the line minimization can no longer detect a decrease, and the line minimization fails although the configuration is essentially relaxed. Summing the pair energies accurately (for example `math.fsum` in Python) and retrying a failed line minimization once along $\mathbf{d} = \mathbf{f}$ both help; if it still fails, use a larger tolerance and state it. A tolerance of 10^{-5} is reachable in all questions of this assignment.
- **Iterations and evaluations.** One CG or SD iteration contains one line minimization, which calls the energy several times. Report both counts.
- **First trial step of the line minimization.** Algorithm 6.3 starts from "some arbitrary step α". Because α has units of length/force, choose it through a length: pick α so that the atom with the largest $\|\mathbf{d}^\alpha\|$ moves by s. Start with $s = 0.01$, then use twice the largest single-atom move of the previous iteration. If $\phi(\alpha_1) \ge \phi(0)$, halve α_1. While $\phi(\alpha_2) \le \phi(\alpha_1)$, double both.
- **CG reset.** Algorithm 6.4 assumes nearly exact line minimizations. With the sufficient-decrease test of Algorithm 6.3, the new direction $\mathbf{d} = \mathbf{f} + \beta \mathbf{d}_{\text{old}}$ can occasionally point uphill ($\mathbf{f} \cdot \mathbf{d} \le 0$). When that happens, set $\mathbf{d} = \mathbf{f}$ and continue.